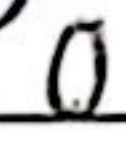


Premolars

- They replace the deciduous molars, so they are "succedaneum" teeth
- All premolars have 4 Lobes (3 buccal + 1 lingual)
except Lower 5 [second premolar] has 5 Lobes
"3 cusp type" (3 buccal + 2 lingual)
- Contact areas of posterior teeth are nearly at the same level and more (cervically)
- Function of premolars :
1st premolars assist canines in tearing of food
2nd premolars assist molars in mastication of food
Lin maintaining vertical dimension of face
support mouth and cheeks
- Thickness > Width (in all premolars)
(BL) (MD)
- Geometric outline, (shortest of even side)
 - * Facial & Lingual $\rightarrow$ Trapezoidal  Cervical
 - * Proximal of:
 - Upper $\rightarrow$ Trapezoidal  Occlusal
 - Lower $\rightarrow$ Rhomboidal 
 - * Occlusal of:
 - U4 $\rightarrow$ Hexagonal 
 - U5 $\rightarrow$ oval 
 - L4 $\rightarrow$ diamond 
 - I 5 $\rightarrow$ rounded or square  

⇒ Maxillary:

[(80% 2 roots) ⇒ U4
 " single root ⇒ U5]

• Contact areas:

- U4 : mesial → in the middle 1/3
 distal $\xrightarrow{\text{more occlusally}}$ at the junction between middle & occlusal 1/3

- U5 : mesial → in the occlusal 1/3
 distal $\xrightarrow{\text{more cervically}}$ at the junction M & O 1/3

• The cusps of premolars

U4 : B cusp > L cusp

U5 : B cusp ≈ L cusp

• U4 has things are opposite for all posterior teeth

→ distal contact area : is located more occlusally (and d. marginal ridge)

→ Deep^{er} d. depression presents on mesial surface not distal [this is the canine fossa]

→ M. cusp slope > d. cusp slope, so the cusp tip is located distally [of the Buccal cusp]

• the occlusal surface of U4 : MB = DB side 
 ML < DL side

• The pulp chamber has 2 pulp horns
under the 2 cusps

⇒ Mandibular

- The maximum convexity of all posterior teeth at lingual surface (in middle $\frac{1}{3}$) expect L5
 In → (occlusal $\frac{1}{3}$)

Lower 1st premolar L4

- has m & d rounded fossae with 2 developmental pits
 ↳ snake eye
- has severe lingual $\angle$ convergence inclination
- Lingual cusp ($\frac{2}{3}$ of length) of buccal cusp
 "non-functioning"
- mesial marginal ridge slopes bucco-lingually
- M. cusp slope $<$ d. cusp slope [of the buccal cusp]

Lower 2nd premolar L5

- mesial marginal ridge is not slopping
- central groove $\rightarrow$ U or H shaped (2 cusp type)
 Y shaped (3 cusp type)
- The 3 cusp type doesn't have lingual convergence and Transverse ridge

The cusps of Lower premolars

L4: B cusp > L cusp

L5: 2 cusp type → B cusp > L cusp

• 3 cusp type → B > ML > DL cusp

Contact areas of Lower premolars

L4: mesial → at the junction M & O_{1/3}
distal → is slightly cervical

Ridges of premolars

Upper: 6 ridges

& L5 (3 cusp type)

- 1 Buccal - 1 cervical

- 2 triangular - 2 marginal

Lower: 7 ridges

(2 cusp type)

- 1 Buccal

- 1 cervical

- 2 triangular

- 1 (transverse)

- 2 marginal

إذا حسبنا الـ (cusp slopes) الـ 5 ridges
اذن نصيب الـ 4 للأعداد السابقة

- mesial marginal developmental groove → in U4
- mesiolingual developmental groove → in L4